



Introduction to Cybersecurity

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Lecture 10

Lecture Rules



Computer Malware and Classification Part II

Lecture Outline

- **Spyware**
- **Worms**
- **Adware**
- **Ransomware**



Spyware

- Spyware is **a software code** or **program**, which is **installed** on the **computers without** letting the **user know** about it.
- The **main objective** of this program is to **monitor online activities, digital marketing campaign**, and **personal interests**.
- Spyware can be **difficult** to **detect**; **often**, the **first indication** a **user** has a **spyware on the system** is a **noticeable reduction** in **processor** or **network connection speeds**.

Spyware Sources

- **File sharing sites** so-called **Torrent** sites is the latest file sharers.
- **Downloading music, movies or software** can pose a **serious risk**.
- **Shareware and Freeware Software**.
- **Unknown emails (Spam emails)**.

Spyware Objectives

- The **objectives** of spreading a Spyware may include the following:
 - **Steal credit card numbers, banking information and passwords.**
 - **Track a person's location.**

Spyware Symptoms

- The **symptoms** of a spyware attack on your computer may include the following:
 - **System** performance **slows down**.
 - **Any program** stops **working properly**.
 - **Many** changes in **browser tools bar** and **plugins appear**.
 - **Advertisements** appear on **your** screen regularly.
 - **Internet bandwidth chokes**.
 - **Search engine** and other **preferences change**.

Spyware Types

- The main categories of the spyware are listed below:
 - **Modem hijacker** spyware.
 - **Browser hijacker** spyware.
 - **Keyboard logger** spyware.
 - **Commercial** spyware.

Countermeasures

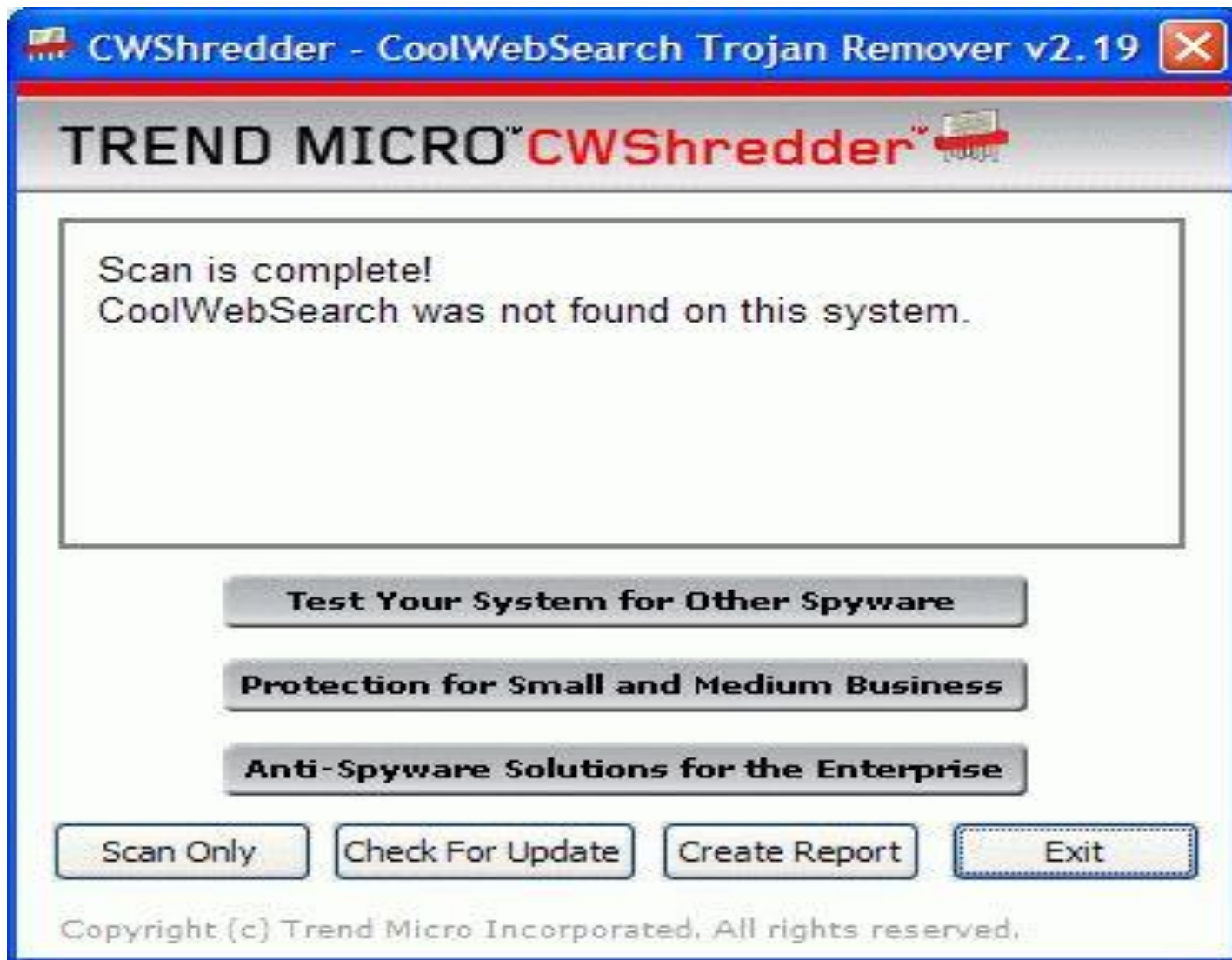
- A few important **measures** to **reduce** the impact of **spyware** are listed below:
 - Always **close** the **advertisement pop-up windows**, Do not click the **OK**, **NO**, **YES**, or **any** other **link** on the **pop-up window**.
 - Be very careful while **visiting new websites** or **redirected websites**.
 - Do not **download free software tools** unless you are **sure** about the **source**.
 - **Activate firewall settings**.
 - **Keep** your **OS** and **anti-spyware** up to date.

Spyware Examples

- **CoolWebSearch (CWS).**
- **Advanced Keylogger.**
- **Internet Optimizer.**
- **GO Keyboard.**
- **Spyware Preventer.**

Spyware Examples

CoolWebSearch (CWS)



Spyware Examples

Advanced Keylogger



Spyware Examples

Internet Optimizer & GO Keyboard



The screenshot shows the TweakBit Internet Optimizer application window. The title bar includes the TweakBit logo, the text "INTERNET OPTIMIZER", and standard window controls. The interface is in Spanish and features a sidebar with navigation options: Inicio, Diagnóstico, Soporte en vivo, Centro de acción (with a red notification badge), and Parámetros. The main content area has a header stating "El rendimiento de su PC y de la conexión pueden ser mejorados". Below this is a speedometer graphic labeled "Velocidad del sistema" with "MIN" and "MAX" markers. To the right of the speedometer is a status box showing "Problemas encontrados: Ninguno", "Último análisis: Hoy", "Versión: Al día", and "Centro de acción: 2 aplicaciones disponibles" with an "instalar todo" link. A large green "Iniciar análisis" button is prominently displayed. Below the button, there is a section titled "¿Por qué activar la versión completa?" with three bullet points: "Arreglar todos los problemas de velocidad en unos instantes", "Programar el mantenimiento automático completo", and "Mantiene a su conexión y a su PC como si fueran nuevas". To the right of these points is a yellow "Activar la versión completa ahora" button. At the bottom left, there is a "¿Necesita ayuda?" section with contact information: "1-888-424-5127" and "Soporte disponible 24*7 (Soporte 24/7 gratis en Estados Unidos y Canadá)". At the bottom right, there is a "GUARANTEED EFFECTIVE 100%" seal and a text box stating "No hay riesgos con nuestra garantía de devolución de dinero de 60 días." and "Llame al 1-888-424-5127 (Soporte 24/7 gratis en Estado Unidos y Canadá)". The bottom of the window shows a "Copias de seguridad" button with a right arrow.

TweakBit
INTERNET OPTIMIZER

ayuda

Inicio

Diagnóstico

Soporte en vivo

Centro de acción

Parámetros

¿Necesita ayuda?

Los expertos pueden ayudarlo en directo con software, hardware u otros problemas del ordenador.

1-888-424-5127

Soporte disponible 24*7 (Soporte 24/7 gratis en Estados Unidos y Canadá)

El rendimiento de su PC y de la conexión pueden ser mejorados

Velocidad del sistema

MIN MAX

Problemas encontrados: Ninguno

Último análisis: Hoy

Versión: Al día

Centro de acción: 2 aplicaciones disponibles [instalar todo](#)

Iniciar análisis

Para conocer el estado actual de su ordenador, [ejecute el análisis de nuevo.](#)

¿Por qué activar la versión completa?

- ✓ Arreglar todos los problemas de velocidad en unos instantes
- ✓ Programar el mantenimiento automático completo
- ✓ Mantiene a su conexión y a su PC como si fueran nuevas

Activar la versión completa ahora

Llame al 1-888-424-5127 (Soporte 24/7 gratis en Estado Unidos y Canadá)

GUARANTEED EFFECTIVE 100%

No hay riesgos con nuestra garantía de devolución de dinero de 60 días.

Copias de seguridad »



The screenshot shows the GO Keyboard application interface on a smartphone. The top status bar displays the time as 11:20 and various system icons. The app's logo, a green leaf with the word "GO" inside, is prominently displayed at the top. Below the logo, the text "GOKeyboard" is written in a large, bold font, followed by the tagline "Great keyboard you can have!". The main part of the screen shows a smartphone with the GO Keyboard app installed. The phone's screen displays a messaging app interface with a list of messages. The GO Keyboard app is shown as a keyboard overlay on the phone's screen. The bottom of the screen shows the standard Android navigation bar with back, home, and recent apps buttons.

11:20

GO

GOKeyboard

Great keyboard you can have!

Go

q w e r t y u i o p

a s d f g h j k l

z x c v b n m

123 En

←

Spyware Examples

Spyware Preventer

Spyware Preventer: WARNING



WARNING! Attack Detected.

Possible harmful action has been detected from remote host.

Spyware Preventer has detected possible harmful actions from remote computer on the network. **Blaster/Sasser, variant** worm behavior detected. You have to register your copy to get full protection feature set and an ability to defeat upcoming threats. To begin online registration, please click "Activate now" button below.

Activate Now!

Block this attack!

Continue unprotected

Disabled protection

Spyware Alert !

REAL TIME
MONITORING
COMPUTER

Vulnerabilities found

Your computer is infected by spyware - 34 serious threats have been found while scanning your files and registry. It is strongly recommended that you disinfect your computer and activate Realtime secure protection against future intrusions.

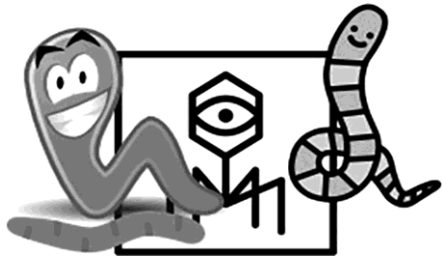
[Why do you need Spyware Protection?](#)



Upgrade to full version of Antivirus System Pro security kit to clean your computer and prevent new security and privacy attacks. You will be able to download daily updates and get online protection against Internet attacks.

Activate Antivirus System Pro

Stay unprotected



Worms

- Worm is a **malware software program** that **replicate** on the **computers** to **consume** the major **parts** of **resources** such as **bandwidth**, **hard disk**, and **memory**.
- The basic **objective** of a **worm** is to **spread** over the **computers** in a **network** by **exploiting** the **vulnerabilities** in the **OS** and **networks**.

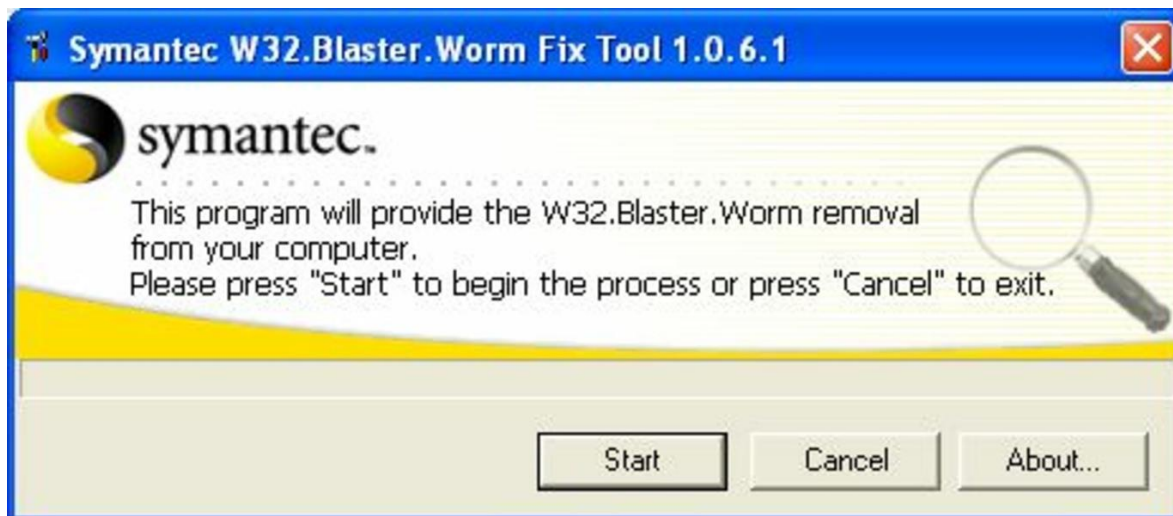
Types of Worms

- A few very important **worms** are listed below:

- **Blaster** worm.
- **MyDoom** worm.
- **Code Red** worm.
- **ILOVEYOU** worm.
- **CIH** worm.

Types of Worms

Blaster worm



Types of Worms

ILOVEYOU worm



Types of Worms



The system is busy waiting for the Close Program dialog box to be displayed. You can wait and click OK, or you can restart your computer.

- * Press any key to return to the command prompt.
- * Press CTRL+ALT+DEL again to restart your computer. You will lose unsaved information in all programs that are running.

Virus.Win9x.CIH

in action in 2023!

Windows

An exception 0E has occurred at 0028:C14953BA in VxD ---. This was called from 0028:C00 in VxD ---. It may be possible to continue normally.

- * Press any key to attempt to continue.
- * Press CTRL+ALT+DEL to restart your computer. You will lose any unsaved information in all applications

Press any key to continue _

Types of Worms



PAYLOAD WORMS



CRYPTOWORMS



BOT WORMS



EMAIL WORMS



DIFFERENT TYPES OF COMPUTER WORMS



**INSTANT MESSAGING
WORMS**



INTERNET WORMS



**FILE SHARING
WORMS**



ETHICAL WORMS

Worms Symptoms

- The main **symptoms** of a worm can be summarized as follows:
 - **Some errors** related to the **OS** and **system files** appear.
 - The **modified** files **disappear** or do **not open**.
 - **Firewall warnings** may **pop-up** in certain cases.
 - **Strange icons** or **files appear** on the desktop.

Worms Symptoms

- The main **symptoms** of a **worm** can be **summarized** as follows:
 - The **performance** of **computer slows** down.
 - **Computer generates** unknown **errors** in **sounds, messages, or even** in **images**.
 - **Sometimes** computer may hang or **freeze**.
 - In **extreme** cases, you will find **emails sent** to your **contacts without** your **knowledge**.

Countermeasures

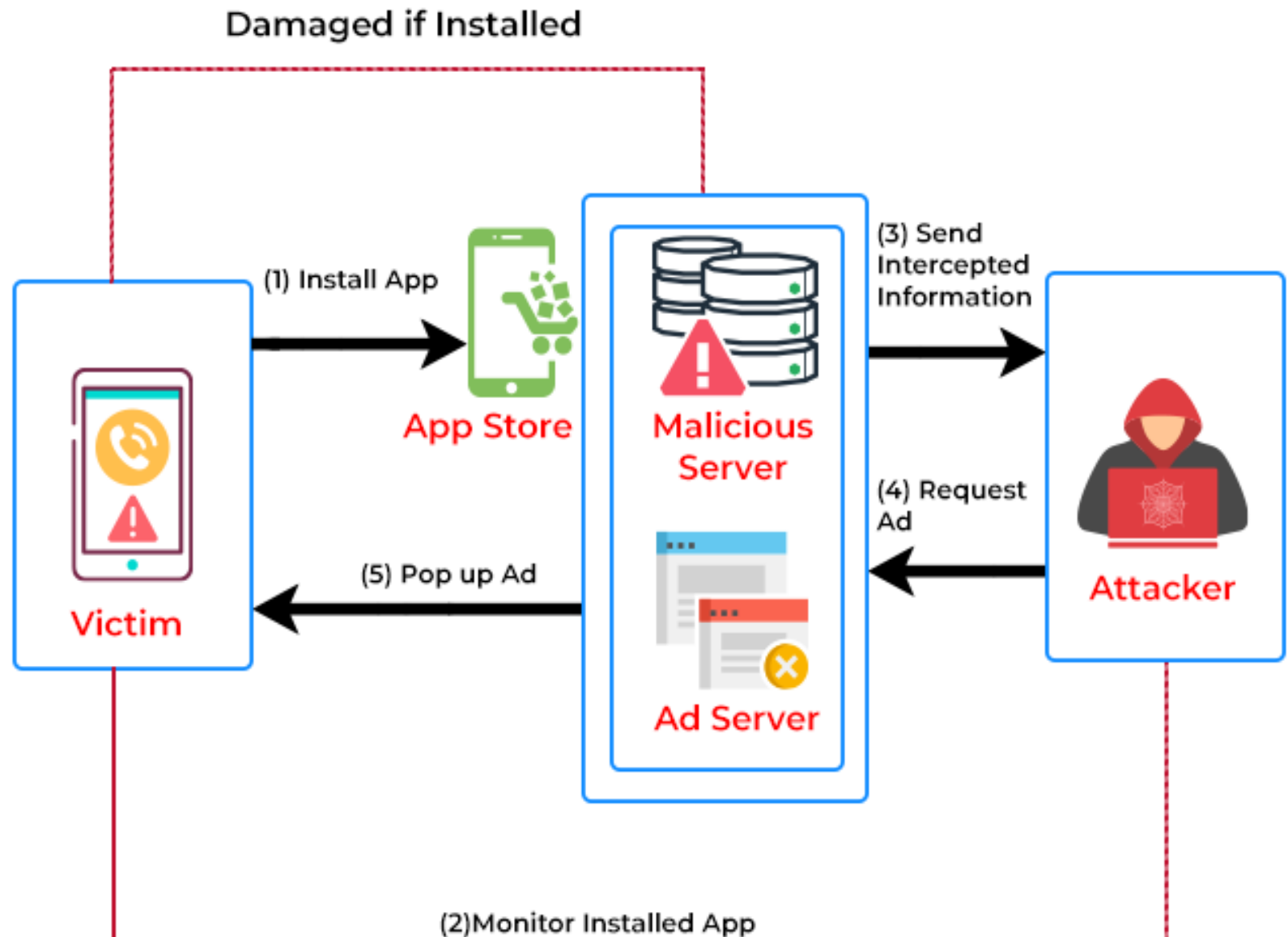
- We can **save our computers** from the **worms** by taking the **following measures**:
 - Always **keep OS updated** with software **patches**.
 - Always take care in **opening emails** and its **attachments** from **unknown source**.
 - **Activate** the **security firewall**.
 - Use **anti-spyware** and **antivirus** software from **trusted** companies.
 - Keep **antivirus software updated**.
 - Use **packet filters** on the **network**.



Adware

- **Adware** is a **computer program** that **forces** the Internet users to visit a **particular web page, pop-up window, or advertisement** to **watch**.
- Adware has **become** a very **popular tool** for **digital marketing teams** to draw attention of the users to a particular **product** or service.

How Adware works?



Adware Symptoms

- A few very important **symptoms** of your computer to have been affected by the **adware** include the following:
 - **Frequent redirects, pop-up** windows of **offers**.
 - **Huge number** of **spam emails**.
 - **Huge number** of **product ads** in **browser**.
 - **Heavy outgoing** and **incoming traffic**.
 - **Slowing down** of **Internet connection**.

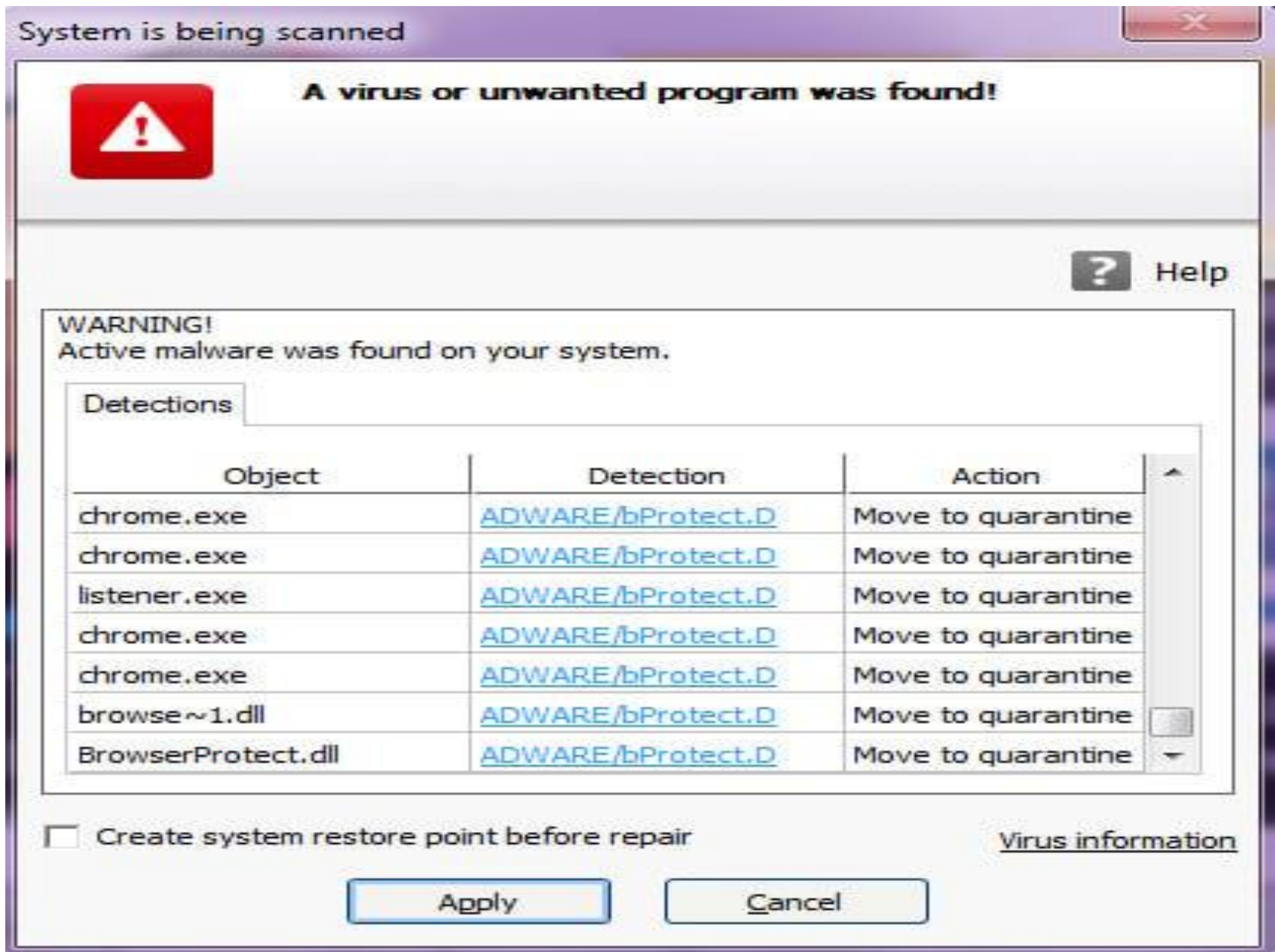
Countermeasures

- You can avoid **adware** or reduce them by taking the following measures:
 - **Activate pop-up blocker** on your **browser**.
 - **Activate** the **firewall**.
 - **Install** and **Update anti-adware software**.
 - Try to **avoid free downloads**, which may **contain adware codes**.

Adware Example



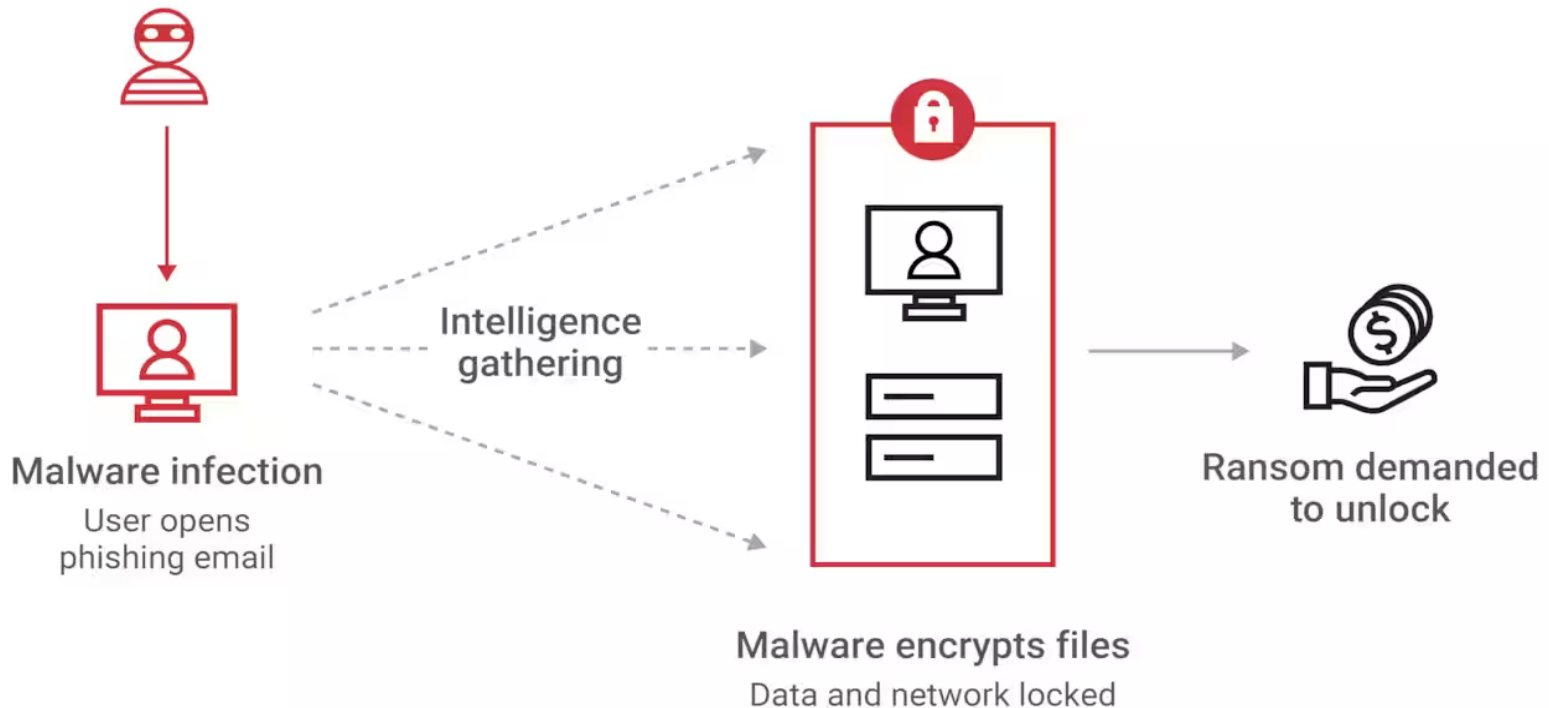
Adware Example





Ransomware

- **Ransomware** is a type of malware that **prevents you from accessing your computer** (or the data that is stored on it). The computer itself may become **locked**, or the **data** on it might be **stolen**, **deleted** or **encrypted**.





Ransomware Symptoms

- A few very important **symptoms** of your computer to have been affected by the **adware** include the following:
 - Encrypted Files (e.g., .locky, .crypt).
 - Pop-up windows, or changed desktop backgrounds.
 - System Lockout.
 - Unusual **Network** Activity.
 - Disabled security software (e.g., antivirus software, firewalls, or other security tools to avoid detection).
 - Slow down **system performance**.

Types of Ransomware



Encrypting Ransomware



Doxware/Leakware



Locker Ransomware



Master Boot Record Ransomware



Scareware



Mobile Device Ransomware

More about ransomware visit this site:

<https://www.tierpoint.com/blog/how-to-avoid-ransomware/>

Examples of Encrypting Ransomware

Extension	Associated Ransomware
.locky	Locky
.crypt	CryptXXX / general use
.zepto	Locky
.cerber	Cerber
.odin	Locky variant
.wallet	Dharma
.crySiS	Crysis/Dharma
.helpme@freespeechmail.org	Crysis
.arena	Dharma
.zzzzz	TeslaCrypt
.vvv	TeslaCrypt
.xtbl	Troldesh / Shade
.exx	GlobeImposter
.mole	CryptoMix
.phobos	Phobos
.makop	Makop
.npsk	STOP/Djvu
.gero	STOP/Djvu
.bora	STOP/Djvu
.djvu	STOP/Djvu
.tor	Onion ransomware



Countermeasures

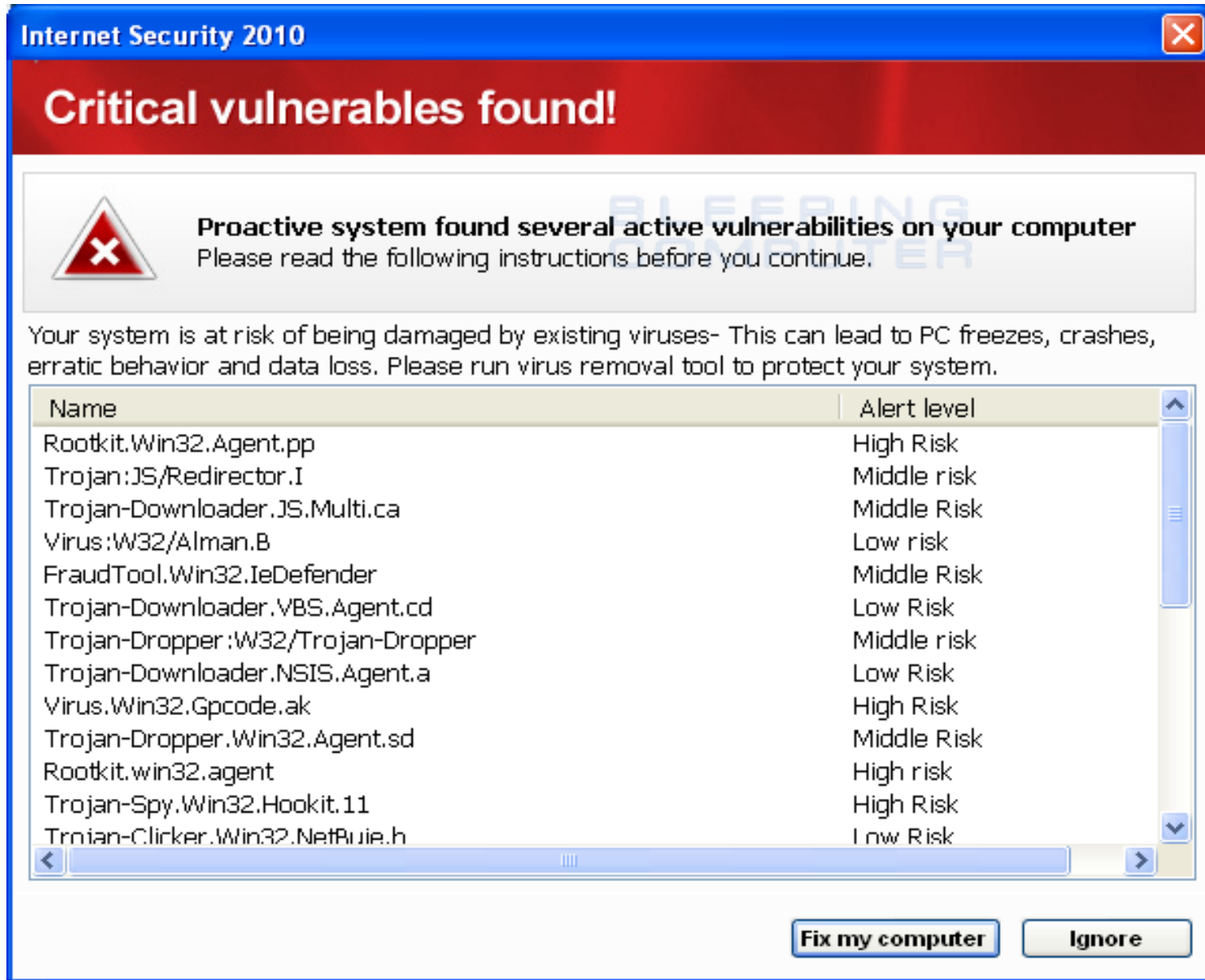
- You can avoid **ransomware** or reduce them by taking the following measures:
 - **Regular** data backups.
 - **Keep OS and software updated.**
 - **Install and update** antivirus **and** anti-malware
 - **Implement** strong **email** security: multi-factor authentication (MFA).
 - **Monitor** Network Activity.
 - **Enable Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS).**



Ransomware Detection



Malware Detection Examples



Questions

Q1. What is the main purpose of a spyware?

Q2. What are differences between Rootkit and Virus?

Q3. List Countermeasures for reducing impact of ransomware?

Q4. List the main types of worms?

Answers

A1:

- The main purpose of spyware is to **secretly monitor and collect user information** without their knowledge or permission. Then, this information is **transmitted** to a third party.

Answers

A2:

ROOTKIT	VIRUS
Rootkit is set of malicious program that enables administrator-level access to a computer network.	A Virus is a malicious .exe code attached to another executable file which can be harmless or can modify or delete data.
It give unauthorized access and control of the system to the attacker.	It can control data and resources, cause error, destroy data and slow down the performance.
It is more harmful.	It is less harmful as compared.

Answers

A3: Countermeasures for reducing impact of ransomware:

- **Regular** data backups.
- **Keep OS and software updated.**
- **Install and update antivirus and anti-malware.**
- **Implement strong email security: multi-factor authentication (MFA).**
- **Monitor Network Activity.**
- **Enable Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS).**

Answers

● **A4:** A few types of **worms** are listed below:

- **Blaster** worm.
- **MyDoom** worm.
- **Code Red** worm.
- **ILOVEYOU** worm.
- **CIH** worm.

Task (20 Min)

1. **Encrypt** the following **Plaintext** using **Affine Cipher** with $K = (5, 22)$.

Attack Amazon Website

2. Given $A=0, B=1, C=2, \dots, Z=25$, **Encrypt** the following **Plaintext** using **Hill Cipher**:
 - **Plaintext: ACT** and **Key: GYBNQKURP**